Prompt workflow

Urs Langenegger

2026/04/21

- Introduction
  - Prompt
  - Reminder on CDB
- Details
- No conclusions

Introduction

- Stages in data processing
  - ▷ "online"
    - what is lost here is gone forever
  - ▷ "prompt"
    - first complete dqm/sort/reco/vtx/ana pass
  - ▷ "(re-)reco"
    - fixing the loose ends of "prompt", legacy for analysis
- "Prompt" comprises
  - ▷ DQM
    - complete statistics
    - "complex" stuff
  - ▷ RDB classification(s) modifications/refinements
  - ▷ "Rolling" (= updated on a run[-block] basis) Calibrations
    - required for optimal reconstruction
    - note that (currently) we do not have differential payloads
  - ▷ Sort/reco/. . .
- Prompt runs workflow suite (many components) on one run
  - ▷ (normally) starts with next run after previous run completed

Reality of life (= Complications)

- Time scale of prompt implies
  - ▷ batch system usage - splitting/merging of jobs
    (without specifying any latency/time scale - yet)
  - ▷ not obvious to what extent Prompt can keep up with data taking
    though in-sequence processing not strictly mandatory
- Probably unstable (= evolving/improving) software
  - ▷ validation of new releases(!)
  - ▷ re-processing of all data accumulated to-date
    - produce new calibration payloads
    - re-run sort/reconstruction/vertexing
  - ▷ run concurrently to normal prompt processing
- Distinguish between
  - ▷ Contents of Prompt
    - what gets executed in which order (below: red = your input required)
  - ▷ Technical setup of Prompt
    - One (big) PERL script in 2025 (it [mostly] worked, imvho)
    - something better for 2026 (including dashboard for monitoring/diagnosis)

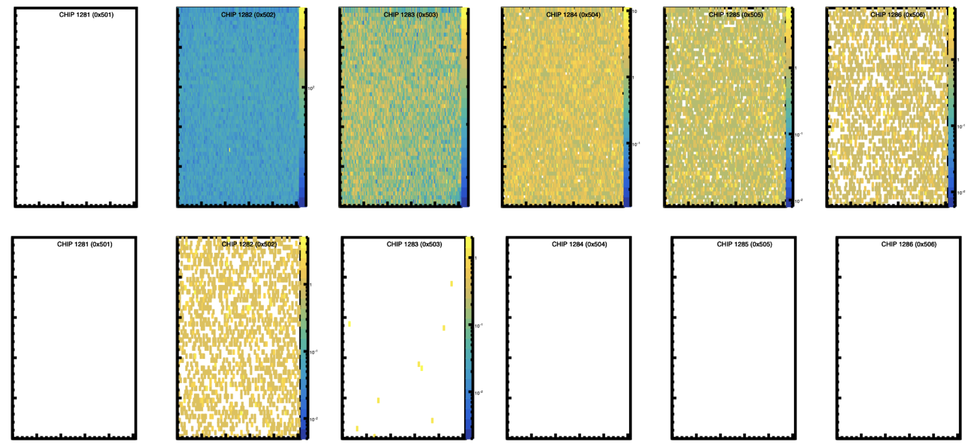
  → Today nothing about technical setup!

Conditions DB? (reminder)

- "Payloads" describe conditions data for analysis of detector data
  - ▷ payloads change if detector conditions change
  - ▷ different implications
    - noise (remove in data)
    - dead (remove in MC)
    - accomodate (in data and MC)
  - ▷ derived in "prompt" workflow (tomorrow)
  - ▷ derived in "offline"

Run 6157:

Run 6164:



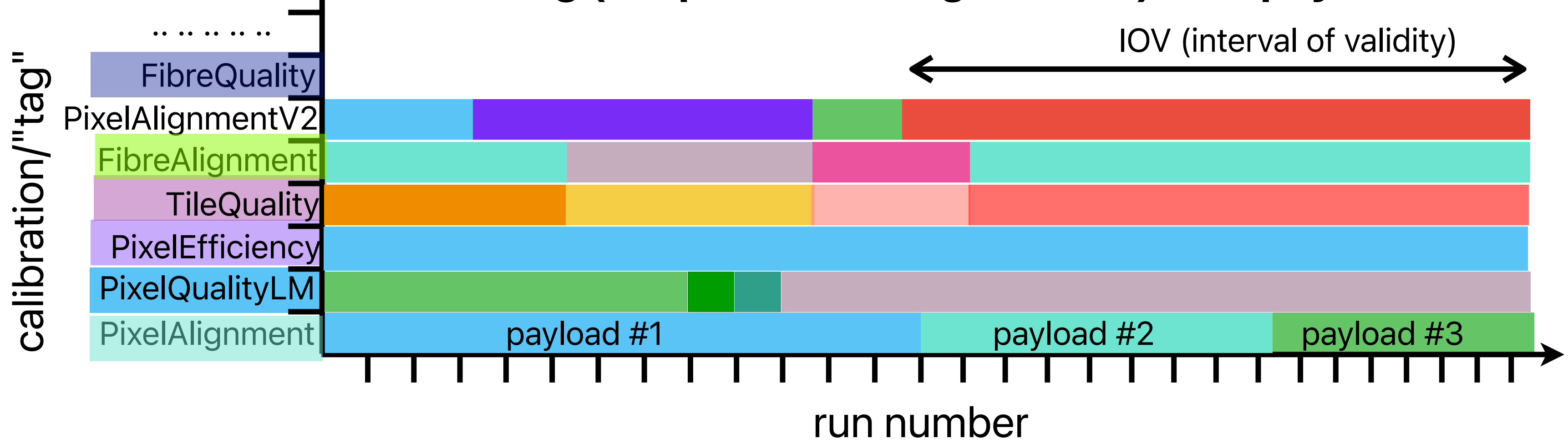
- Payloads belong to certain "calibrations" (e.g. pixelQualityLM)
  - ▷ they come in different versions
    improvements (include MIDAS meta data)
    bugfixes (include MIDAS meta data correctly)
    more alignment analyses
  - ▷ they apply to certain run ranges (iov = interval of validity)
    can change with versions

```
[moor>cat ~/data/mu3e/cdb/tags/detsetupv1_datav6.3=2025V0
  { "tag" : "detsetupv1_datav6.3=2025V0", "iovs" : [ 1, 2177, 6302 ] }
```

→ there are (nearly) always multiple payloads for a calibration
"vector" of iovs

Global tags, tags, IOVs, payloads

Global tag (simplified naming scheme) with payloads



Global tag with "real" names

datav6.3=2025V1

pixelalignment_datav6.3=2025V1
fibrealignment_datav6.3=2025V0
mppcalignment_datav6.3=2025V0
tilealignment_datav6.3=2025V0
pixelqualitylm_datav6.3=2025V0
detsetupv1_datav6.3=2025V0
pixeltimecalibration_datav6.3=2025V0
pixelefficiency_datav6.3=2025V0
tilequality_datav6.3=2025V0
fibrequality_datav6.3=2025V0
eventstuff_datav6.3=2025V1

- Terminology
  - payload - conditions data describing detector status (conditions)
  - IOV - interval of validity (open ended: only start provided)
  - tag - name identifying "calibration", together with vector of IOVs
  - global tag (GT) - complete set of tags

Detailed Prompt workflow I

- Prerequisites
  - ▷ dedicated partition on merlin6 batch system
    - not realistic to share this with (analysis) users during data taking
  - ▷ Dedicated diskspace on merlin6
  - → should all be fine, agreement with merlin6 admins/Nik+Stefan
  - ▷ automation!
- Initialization of Prompt release
  - ▷ "tagged" release
    - mu3e
    - minalyzer
    - auxiliary code (payload determination, plotting, . . .)

```
[login001>l /data/project/mu3e/offline/260421
total 17
drwxrws---+ 2 langenegger svc-cluster_mu3e 4096 Apr 21 10:52 mu3e
drwxrws---+ 2 langenegger svc-cluster_mu3e 4096 Apr 21 10:52 minalyzer
drwxrws---+ 2 langenegger svc-cluster_mu3e 4096 Apr 21 10:53 anaComba
drwxrws---+ 2 langenegger svc-cluster_mu3e 4096 Apr 21 10:53 dqm-analysis
-rw-rw----+ 1 langenegger svc-cluster_mu3e  599 Apr 21 10:53 githashes
```

- Setup per run
  - ▷ check for new run data arrival
  - ▷ check whether it should be processed
    - run category?
    - number of events!?
  - ▷ run mu3e_midas_meta

Prompt workflow II

- minalyzer
  - same code base as online (could potentially include more histograms)
  - run on complete statistics in merlin6 batch
    - split into blocks
    - combine output after all jobs finished successfully (cope with crashes)
  - validate output
- DQM output
  - run some ROOT (compiled:) code (displaying/fitting/rebinning/. . .)
    - 2025: pdf
    - 2026: whatever else in addition you want
  - upload pdf to RDB
  - transfer root file to mu3ebe (for "online" DQM/custom page)
- CDB payloads
  - based on minalyzer output and midas metadata (and more, see below)
    - 2025: pixelQualityLM from compiled code
    - 2026: all other (interested) subsystems
    - validation of payloads
    - update CDB on mu3edb0 (payload, add runnumber to tag IOV array, validate)

Prompt workflow III

- These steps require successful completion of previous step
  - ▷ CDB update (because steps below pull payloads from CDB)
- mu3eSort
  - ▷ no splitting into blocks(?)
  - ▷ configuration(?)
  - ▷ calibrations to be applied?
    - pixelQualityLM: links? columns? pixel? keep "suspicious"?
    - timing
    - fibres, tiles, . . .

tiles calibration after sort

- mu3eTrirec
  - ▷ with splitting into blocks
  - ▷ calibrations to be applied?
    - pixelQualityLM: links? columns? pixel? keep "suspicious"?
    - timing, fibres, tiles, . . .
  - ▷ combine output
- vertexing/. . .

Validation requirements

- Each calibration must provide validation output
  - ▷ technical
    - code ran successfully to completion
    - calibration constants are valid (e.g., non-zero)
  - ▷ calibration constants summary
    - determine run-quality from (e.g.) *quality

  → essential subsystem/DQM input!

  how many "bad" fibres?

  how many de-sync'ed links?

  . . .
- Imvho do not run code that does not return validation output

Thoughts on integrating code

- Prompt workflow relies on many "additional" components
  - ▷ 2025
    - minalyzer
    - Mark Wong's code
    - anaComba (attempted, but stopped at some point)
  - ▷ 2026
    - quality determination for fibres, tiles, links, . . .

  → clone "tagged" repository from anywhere (github, bitbucket, gitlab, . . .)
- Technical wishes
  - ▷ it should compile on merlin(6)
  - ▷ example how to call the code and what output it produces
  - ▷ example how to interpret the validation output
- CDB payloads
  - ▷ if output (json, csv, xml) stays as 2025, then I have code to do it
  - ▷ generally I am happy to provide the translation from your output to payloads
  - ▷ Note: I will very strongly (try to) object to schema changes during run time!
    (i.e. no adding of additional information to a payload)